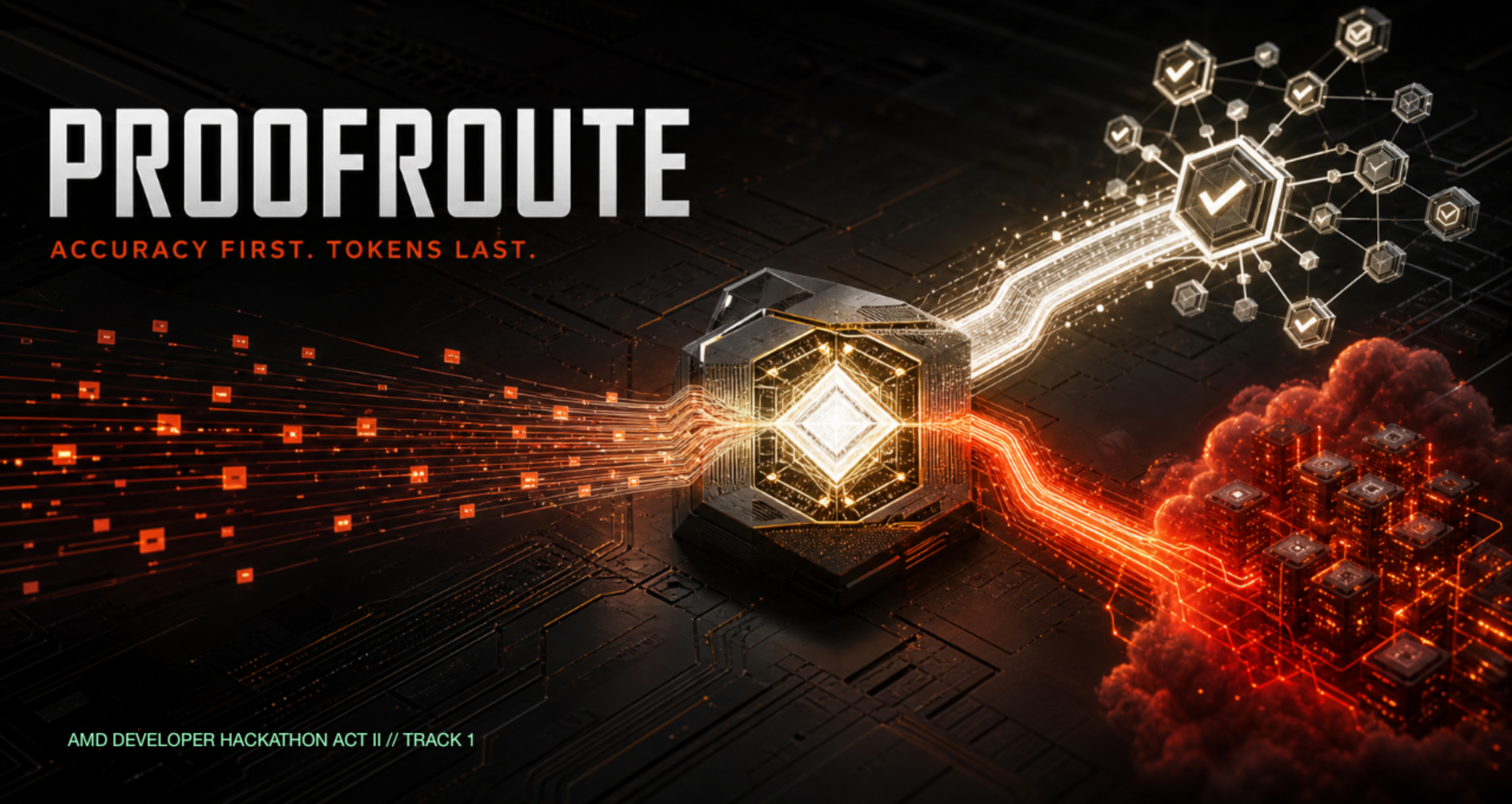


PROOFROUTE

ACCURACY FIRST. TOKENS LAST.



AMD DEVELOPER HACKATHON ACT II // TRACK 1

Not every query should pay the full price

ProofRoute turns inference cost into a decision.

Prove deterministic work locally. Use calibrated local intelligence only where evidence supports it. Buy remote intelligence when uncertainty demands it.

Accuracy is the gate. Token efficiency is the advantage.

ALWAYS REMOTE

FULL TOKEN COST

PROOFROUTE

LOCAL

REMOTE ONLY WHEN
NEEDED

Two evidence-backed paths share one answer contract

SUBMITTED RUNTIME



CALIBRATED LOCAL CHALLENGER



The engine—not the model—owns JSON structure. Raw-prompt ablation cut Fireworks input tokens by 51.9% with byte-identical Kimi answers.

Deterministic answers require proof, not confidence

- 01** Parse a registered contract
bounded math, logic, extraction or code
- 02** Recompute the candidate
safe AST, graph closure or constrained execution
- 03** Require one unique result
ambiguity, mismatch or unsupported input refuses
- 04** Apply the answer contract
release zero Fireworks tokens or fall through

2 / 2

known correct ordering candidates
recovered

5 / 5

incorrect deterministic candidates refused

Independent gate: 180 releases • 100%
precision • 0 false positives

AMD training converts task signals into selective local routing

FunctionGemma learns the task. Gemma E2B attempts the answer. Judges label the outcome. Regression learns the frontier.

DATASET

4,000 questions

8 balanced categories

FUNCTIONGEMMA

270M parameters

intent + 5 scores

GEMMA 4 E2B

96-token ceiling

post-contract answer

ADJUDICATION

Fireworks +
Gemini

correct / incorrect

REGRESSION

per intent

local or remote

AMD ROCm 7.2 • PyTorch 2.9 • Q8 FunctionGemma • text-only LiteRT E2B

3,982 valid task/parameter intersections; 18 invalid 270M assessments fail closed to Fireworks.

Measured precision makes the token frontier actionable

828

correct E2B answers after Answer
Contract Engine

84.52%

out-of-fold precision • per-intent
regression

12.66%

selective local coverage • 252 of 1,991

THE PUBLIC RELEASE

45.7 MB

compressed linux/amd64 image

595 TESTS

CI + Docker gates green

1 SECOND

official smoke runtime

More answers per token. Accuracy protected.